

Curriculum Based Measurement in Reading (CBM-R)

Purpose, use, and administration guidelines

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Purpose of CBM-R

- Curriculum Based Measurement aims to index the level and rate of student achievement in reading.
- Monitor intervention effectiveness
- CBM is a General Outcome Measure (GOM), which
  - Indexes long-term growth in the reading curriculum;
  - Measures oral reading fluency across time;
  - Monitors progress.

Purpose of CBM-R

CBM-R can be used for Data-Based Decision Making[1,2] when used as a measure of:

- ***Screening***
  - Compare students to local grade level norms at the classroom, school, and district level.
- ***Benchmarking***
- ***Progress monitoring***
  - To establish baseline;
  - To measure growth overtime using standardized procedures.



CBM-R Research

- CBM-R provides the capacity to model reading growth:
  - Several theoretical models of reading frequently conceptualize oral reading fluency as an indicator of overall reading achievement. [3, 5]
  - Reading fluency within the context of passage is a strong predictor of comprehension. [4]



Overview of CBM-R Procedures

- Student reads from passage while errors are recorded.
- Each administration takes one minute.
- Passages are approximately 200-300 words.
- Typically administered during grades 1-5.
- Passages are created from meaningful, connected text.
- Recommended to administer three passages and take the median score.



Overview of CBM-R Procedures: Following Standardized Procedures

- Must administer the test the same way with each administration.
- Standardized procedures:
  - Directions
  - Scoring procedures
- It is also recommended to use standardized reading passages.



Administering CBM-R

- In order for CBM-R administration to be reliable and accurate, it is important to make sure that:
  - All directions are followed exactly.
  - All errors are marked appropriately.
  - The last word read is marked appropriately.
  - No errors are corrected.
  - No feedback is given.



Administering CBM-R

Materials

- Passage
  - Student copy
  - Administrator copy (numbered)
- Pen/pencil
- Clipboard
- Timer/Stopwatch

Directions

- The same directions should be given for each administration.
- Several versions exist for delivering directions to students.



Directions

For each passage in the session, use the following directions

***Today you are going to read three different stories out loud to me. I will give you directions for each story. Do* you have any general questions? OK then, let's begin.**

"This is a story about _____ (&_____)."
(*Point to the names or place name at the top of the story;*

"When I say 'BEGIN,' start reading aloud at the top of this page. Read across the page."
(Point to the first word and across the first line).

"Try to read each word. If you come to a word you don't know, I'll tell it to you. <u>Be sure to do your BEST reading</u>.

OK?"
(Pause for response)

"Remember, you're doing your BEST reading."

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Alternate Directions

Example 1

"I want you to read this story to me. You'll have 1 minute to read.

When I say, 'begin,' start reading aloud at the top of the page.

Do your best reading.

If you have trouble with a word, I'll tell it to you.

Do you have any questions? Begin."

(National Center for Response to Intervention)

Example 2

"Please read this (point) **out loud.**

If you get stuck, I will tell you the word so you can keep reading.

When I say, "stop," I may ask you to tell me about what you read, so do your best reading.

Start here (point to the first word of the passage)**.**

Begin."

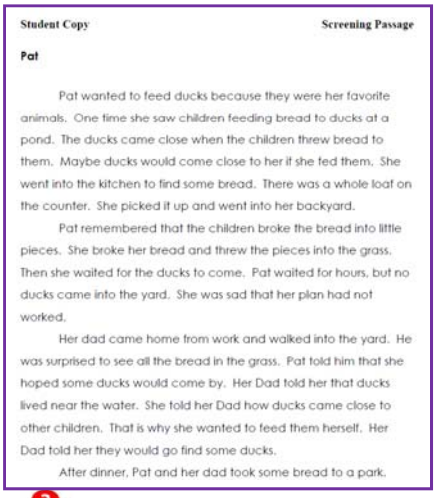
(DIBELS)

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Student and Administrator Passages

- Notice the administrator copy includes a cumulative word count.



Student Copy **Screening Passage**

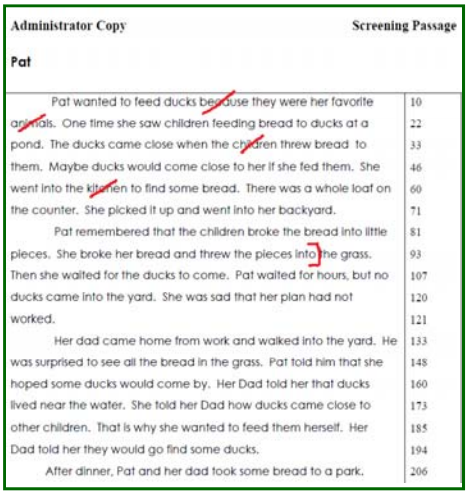
Pat

Pat wanted to feed ducks because they were her favorite animals. One time she saw children feeding bread to ducks at a pond. The ducks came close when the children threw bread to them. Maybe ducks would come close to her if she fed them. She went into the kitchen to find some bread. There was a whole loaf on the counter. She picked it up and went into her backyard.

Pat remembered that the children broke the bread into little pieces. She broke her bread and threw the pieces into the grass. Then she waited for the ducks to come. Pat waited for hours, but no ducks came into the yard. She was sad that her plan had not worked.

Her dad came home from work and walked into the yard. He was surprised to see all the bread in the grass. Pat told him that she hoped some ducks would come by. Her Dad told her that ducks lived near the water. She told her Dad how ducks came close to other children. That is why she wanted to feed them herself. Her Dad told her they would go find some ducks.

After dinner, Pat and her dad took some bread to a park.



Administrator Copy **Screening Passage**

Pat

Pat wanted to feed ducks because they were her favorite	10
animals. One time she saw children feeding bread to ducks at a	22
pond. The ducks came close when the children threw bread to	33
them. Maybe ducks would come close to her if she fed them. She	46
went into the kitchen to find some bread. There was a whole loaf on	60
the counter. She picked it up and went into her backyard.	71
Pat remembered that the children broke the bread into little	81
pieces. She broke her bread and threw the pieces into the grass.	93
Then she waited for the ducks to come. Pat waited for hours, but no	107
ducks came into the yard. She was sad that her plan had not	120
worked.	121
Her dad came home from work and walked into the yard. He	133
was surprised to see all the bread in the grass. Pat told him that she	148
hoped some ducks would come by. Her Dad told her that ducks	160
lived near the water. She told her Dad how ducks came close to	173
other children. That is why she wanted to feed them herself. Her	185
Dad told her they would go find some ducks.	194
After dinner, Pat and her dad took some bread to a park.	206

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Scoring CBM-R

Score Immediately

- Determine words read correctly (WRC) / min.
- Total words read - Errors = WRC / min
- Score is WRC / Errors

Example:

- "Ted left his burlcase at home."
- 6 (TWR) – 1 (Error) = 5 WRC / min

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Scoring CBM-R: What is an Error?

Mispronunciation of a word

- "Instead, she would ***read*** books from the library."
- Pronounced *'red'*, instead of *'read'*.

Word Substitutions

- The sentence is written:
  - "She ***hoped*** to compete in the horse show at the fair."
- The sentence is read:
  - "She ***hopped*** to compete in the horse show at the fair."

Three second pauses or struggles



Scoring CBM-R: What is an Error?

Word Reversals *(two words)*

- The sentence is written:
  - "***Now Anna*** knew where the stream began and ended."
- The sentence is read:
  - "***Anna now*** knew where the stream ended and began."

Omissions

- The sentence is written:
  - "She tried to do the same ***things that he*** did."
- The sentence is read:
  - "She tried to do the same ***things he*** did."



Scoring CBM-R: What is NOT a WRC or an Error (and CAN Be Counted)?

Repetitions

- –The sentence is written:
  - "Her father said ***that*** some horses do not like to jump."
- –The sentence is read:
  - "Her father said ***that … that*** some horses do not like to jump."

Dialect Differences

- –"It was much too small ***for*** her to wear now."
  - Pronounced *'fur'*, instead of *'for'*.

Insertions

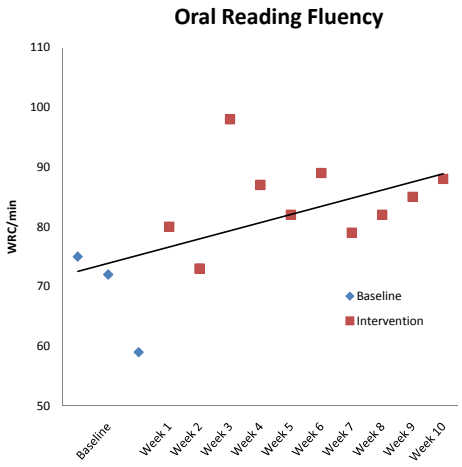
- –The sentence is written:
  - "Sally was the youngest alligator in her family."
- –The sentence is read:
  - "Sally was the youngest alligator in ***all of*** her family."

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Graphing CBM-R

1. Establish baseline
2. Set student goal
3. Progress monitor
4. Decision rules



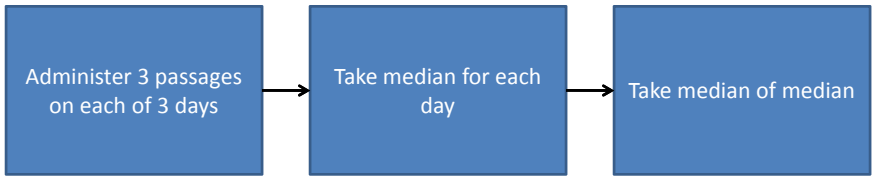
Note: Points should not be connected.

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Graphing CBM-R:Procedures

1. Establish Baseline



WRC/min

Day 1	Day 2	Day 3
72*	**73***	69
68	78	**71***
75	71	73

= **Baseline 72**

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Graphing CBM-R: Procedures

2. Goal should be ambitious, yet attainable.
   - Define expected rate of growth
     - Typically set at 1-2 WRC/min per week
3. Establish Progress Monitoring Schedule
   - Define number of weeks

Goal = (Expected Rate of Growth x Weeks) + Baseline

Example:

- Student Baseline Performance = 72 WRC/min
- Expected Growth Rate = 1.5
- Weeks = 10

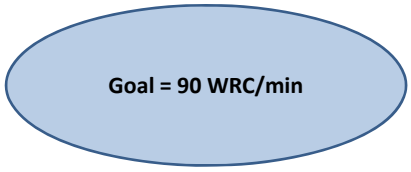
→ **(1.5 × 10) + 72**
Goal = 87 WRC/min

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Goal Setting Alternative*

1. Using at least 8 data points determine rate of weekly improvement using slope.
   - Slope = 1.2
2. Multiply slope by the average rate of growth (1.5 WRC/min)
   - 1.2 × 1.5 = 1.8
3. Multiply by number of weeks of intervention
   - 1.8 × 10 = 18
4. Add to baseline
   - 18 + 72 = 90



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* National Center for Progress Monitoring

Decision Rules

- Data point decision rules help guide the decision making process

If (3,4,5) most recent data points are:

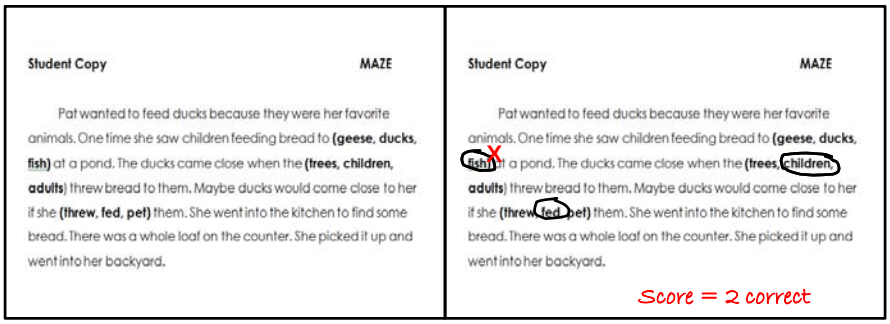
- ABOVE the line → keep current intervention, increase goal.
- BELOW the line → keep current goal, modify instruction.
- Neutral – points are above and below the line → maintain current goal and instruction .

National Center for Progress Monitoring recommends 5 data points.

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CBM MAZE

- Student circles correct word out of 3 choices.
- Administration lasts 3 minutes.
- Score is the number of correct words.



Student Copy MAZE

Pat wanted to feed ducks because they were her favorite animals. One time she saw children feeding bread to **(geese, ducks, fish)** at a pond. The ducks came close when the **(trees, children, adults)** threw bread to them. Maybe ducks would come close to her if she **(threw, fed, pet)** them. She went into the kitchen to find some bread. There was a whole loaf on the counter. She picked it up and went into her backyard.

Student Copy MAZE

Pat wanted to feed ducks because they were her favorite animals. One time she saw children feeding bread to **(geese, ducks, fish)** at a pond. The ducks came close when the **(trees, children, adults)** threw bread to them. Maybe ducks would come close to her if she **(threw, fed, pet)** them. She went into the kitchen to find some bread. There was a whole loaf on the counter. She picked it up and went into her backyard.

Score = 2 correct

Summary

- Administration of a one-minute passage, recording errors.
- Always use the ***standardized directions.***
- Students should not practice reading the standardized passages ***(not instruction).***
- Timing procedures should be subtle.
- This is ***NOT an assessment of speed reading***, but an assessment of ***BEST*** reading.
- Discard passage and administer another if disrupted.

Resources

DIBELS
- https://dibels.uoregon.edu/

AIMSweb
- www.aimsweb.com

Intervention Central
- http://www.interventioncentral.org/

National Center on Response to Intervention
- http://www.rti4success.org/

References

1. Deno, S. L. (1986). Formative evaluation of individual student programs: A new role for school psychologists. *School Psychology Review, 15*(3), 358-374.
2. Deno, S. L. (2003). Developments in Curriculum-Based Measurement. *Journal of Special Education, 37*(3), 184-192.
3. Fuchs, L. S., Fuchs, D., Hosp, M. K., & Jenkins, J. R. (2001). Oral reading fluency as an indicator of reading competence: A theoretical, empirical, and historical analysis. *Scientific Studies of Reading, 5*(3), 239-256.
4. Hosp, M & Fuchs, L. (2005) Using CBM as an indicator of decoding, word reading, and comprehension: Do the relations change with grade? *School Psychology Review, 34*, 9-26.
5. Shinn, M. R., & Others, A. (1992). Curriculum-based measurement of oral reading fluency: A confirmatory analysis of its relation to reading. *School Psychology Review, 21*(3), 459-479.

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Additional Training Modules

- Using CBM for Progress Monitoring in Reading (National Center on RTI)
- Classroom Assessment (Part 2): Evaluating Reading Progress: (Peabody Iris Center)
- Using CBM for Progress Monitoring in Reading (National Center on Student Progress Monitoring

